1. An air-track glider of mass 150 g is attached to a spring with a spring constant of 0.95 N/m. At $t = 0$ s the glider is 4.40 cm to the right of the equilibrium position (x=0) and is moving to the left at 6.3 cm/s. What is the amplitude of the motion?

a. 16.2 cm b. 13.5 cm c. 5.06 cm d. 20.7 cm e. other (give the value)

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.95}{150 \times 10^{-3}}} = 2.52$ $x = A\cos(\omega t + \phi)$ $v = \frac{dx}{dt} = -A\omega \sin$

$\frac{dv}{dt} = -A\omega^2 \cos(\omega t + \phi)$

$\frac{1}{2}kA^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$

$\frac{1}{2}(0.95)A^2 = \frac{1}{2}(0.95)(0.044)^2 + \frac{1}{2}\left(\frac{150}{1000}\right)(0.063)^2$

$A = 0.0506 \text{ m} = 5.06 \text{ cm}$

2. The speed of an object vibrating along a horizontal direction obeys the equation $v(t) = 0.445\sin(25.4t + 0.223)$ where the speed is in m/s and time in seconds. What is the acceleration of the vibrating object at t=2 seconds?

a. 8.2 m/s^2 b. 0.32 m/s^2 c. 0.45 m/s^2 d. 11.0 m/s^2 e. other (give the value)

$v(t) = 0.445\sin(25.4t + 0.223)$

$\frac{dv}{dt} = a(t) = 0.445(25.4)\cos(25.4t + 0.223)$

$a(2) = 0.445(25.4)\cos(25.4(2) + 0.223)$

$a \doteq 8.2 \frac{m}{s^2}$

3. The motion of a spring-mass system is described by $x(t) = 2\cos(\omega t + \phi)$ cm. At t=0 s, the position of the mass is x=1 cm and the oscillating velocity is positive. What is the phase constant ϕ in radians?

a. $\pi/2$ b. $-\pi/3$ c. $-\pi/2$ d. $\pi/3$ e. $\pi/6$

$x(0) = 1 = 2\cos(\omega(0) + \phi)$

$\cos(0 + \phi) = \frac{1}{2}$

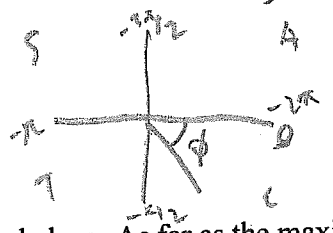
$\phi = \frac{\pi}{3}$

$x(t) = 2\cos(\omega t + \phi)$

$\frac{dx}{dt} = v(t) = -2\omega \sin(\omega t + \phi)$

$v(0) > 0$

$\therefore \sin(\omega t + \phi) < 0$

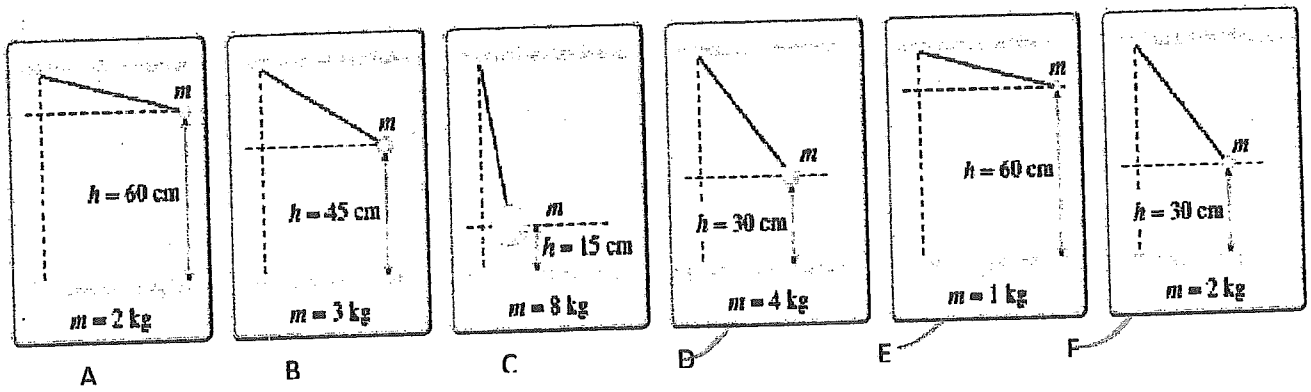


$\phi = -\frac{\pi}{3} \rightarrow \sin\theta < 0$, $\cos\theta > 0$

4. Observe the figure given below. As far as the maximum speed of each pendulum is concerned, they are ranked, from larger to smaller, as:

a. B>A=C=D>E=F b. A>B>C=D=E>F c. C>D>B>A=F>E

d. A=E>B>D=F>C e. None of the above



$E_T = E_T'$

$mgh = \frac{1}{2}mv^2$

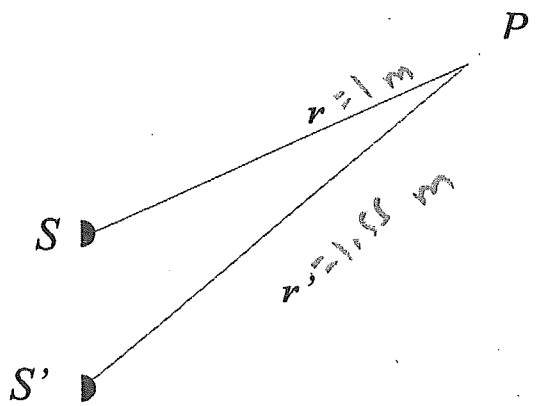
$v = \sqrt{2gh}$ $\therefore$ h increases

v increases

5. Two coherent sound sources S and S' emit in phase sound waves of wavelength 10 cm. The distances of point P to the sources S and S' are r=1.00 m and r'=1.55 m respectively. If the pressure amplitude of each wave at point P is 2.00 Pa, what is the total pressure amplitude at point P (in Pa)?

a. 1.6 b. 0 c. 2.0 d. 4.0 e. other (give value)

$\frac{|\Delta r|}{\lambda} = \frac{\phi}{2\pi}$



$\frac{|1.55-1|}{0.1} = \frac{\phi}{2\pi}$

$\phi = 11\pi$

$A = 2A\cos\left(\frac{\phi}{2}\right)$

$A = 2(2)\cos\left(\frac{11\pi}{2}\right) = 0$

6. Two ropes are spliced together as shown and a pulse is traveling to the right on them.

A short time after the incident pulse shown in the diagram above, reaches the splice, what will be the rope's appearance?

a.

b.

c.

d.

e.

7. If a wave is described by $y(x,t) = 0.02 \sin(30x - 400t)$ (SI units), the wavelength of the wave is

a. $\pi/15$ m
b. $15/\pi$ m
c. 60π m
d. 4.2 m
e. 30 m

$y(x,t) = A\sin(kx - \omega t)$

$k = 30$

$\lambda = \frac{2\pi}{k} = \frac{2\pi}{30} = \frac{\pi}{15}$ m

8. A transverse wave in a string is described by $y(x,t) = 0.15\sin(\frac{\pi}{16}(2x - 64t)$ where x and y are in meters and t in seconds. Determine the tension in the string if its linear mass density is 3.0×10^{-4} kg/m.

$v = \sqrt{\frac{T}{\mu}}$ $\mu = 3\times10^{-4}\ \frac{kg}{m}$

a. 0.074 N
b. 0.156 N
c. 2.47 N
d. 0.307 N
e. other (give value)

$v = \lambda f = \frac{2\pi}{k} \cdot \frac{\omega}{2\pi} = \frac{\omega}{k} = \frac{\frac{64\pi}{16}}{\frac{2\pi}{16}}$

$\frac{64\pi}{16} \times \frac{16}{2\pi} = 32$ m/s

$32 = \sqrt{\frac{T}{3\times10^{-4}}}$

$T = 0.307$ N

9. A wave generated in a medium is a longitudinal wave when

a. there is a net transport of matter by the wave. X

b. the molecules of the medium are unable to exert forces on each other.

c. molecular displacements are parallel to the wave velocity. ✓

d. molecular displacements are perpendicular to the wave velocity. X

e. the density of the medium is less than the density of water.

10. As a certain sound wave travels through some gas of density 1.40 kg/m^3, the displacement wave function is given by $s(x, t) = 2.54\times10^{-6} \sin(\pi x/2 + 200\pi t)$ in SI units. What is the pressure amplitude?

s_{max} ω

a. 0.14 N/m^2 b. 0.28 N/m^2 c. 0.65 N/m^2 d. 0.76 N/m^2 e. 0.89 N/m^2

$\Delta P_{max} = \rho v \omega s_{max}$

$v = \lambda f = \frac{2\pi}{k} \cdot \frac{\omega}{2\pi} = \frac{\omega}{k} = \frac{200\pi}{\pi/2}$

$= 200\pi \cdot \frac{2}{\pi}$

$\rho = 1.4 \frac{kg}{m^3}$

$v = 400 \frac{m}{s}$

$\omega = 200\pi \frac{rad}{s}$

$= (1.4)(400)(200\pi)(2.54\times10^{-6})$

$\Delta P_{max} \doteq 0.89 \ N/m^2$

11. A pipe has a diameter of 1.0 cm and a length of 80 cm. It is open at both ends and immersed in a gas mixture (not air!). Standing sound waves are set up along its length, such that the frequency of the *n*-th harmonic is 280 Hz. The next higher harmonic has a frequency of 350 Hz. Determine the number *n*.

a. 2 b. 3 c. 4 d. 5 e. other (give value)

$f = \frac{nv}{2L}$

$280 = \frac{nv}{2(0.8)}$

$448 = nv$

$v = \frac{448}{n}$

$350 = \frac{(n+1)v}{2(0.8)}$

$560 = (n+1)v$

$560 = (n+1)\frac{448}{n}$

$\frac{560}{448} = \frac{1}{n}(n+1)$

$1.25 = 1 + \frac{1}{n}$

$0.25 = \frac{1}{n}$

$n = \frac{1}{0.25}$

$n = 4$

12. On a winter day, when the air temperature is $0^\circ C$, a truck travels northward on a highway at 30.0 m/s. A police car, also traveling northward, approaches the truck at 40.0 m/s from behind, with its siren sounding a frequency of 2000 Hz. What frequency does the truck driver detect before the police car overtakes him?

a. 2239 Hz

b. 1635 Hz

c. 2946 Hz

d. 2069 Hz

e. other (give value)

$V_{sound} = V = 331\sqrt{1 + \frac{T_c}{273}} = 331\sqrt{1 + \frac{0}{273}} = 331 \frac{m}{s}$

$f' = f\left(\frac{V - V_0}{V - V_s}\right)$

$V_0 = 30$ m/s

$V_s = 40 \frac{m}{s}$

$f = 2000$ Hz

$= 2000\left(\frac{331 - 30}{331 - 40}\right)$

$= 2000\left(\frac{301}{291}\right)$

$\doteq 2068.7$ Hz

$f' \doteq 2069$ Hz